

PHD ANALYSIS EXAM, AUGUST 2019

DO THREE FROM PART A AND THREE FROM PART B. ANSWER EACH PROBLEM ON A SEPARATE SHEET OF PAPER. WRITE SOLUTIONS IN A NEAT AND LOGICAL FASHION, GIVING COMPLETE REASONS FOR ALL STEPS.

**Part A**

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- (1) Let  $X, Y$  be topological spaces. Prove that every continuous function  $f : X \rightarrow Y$  is Borel measurable.
- (2) Short answer.
- (a) Give an example, if possible, of a *closed* set  $E \subset \mathbb{R}$  of positive Lebesgue measure that contains no interval.
- (b) For  $n \geq 2$ , let  $f_n = e^{inx} x^{-n}$ . Find

$$\lim_{n \rightarrow \infty} \int_1^\infty f_n dx.$$

- (c) Give an example, if possible, of strict inequality in Fatou's Lemma.
- (3) Let  $X = Y = [0, 1]$ ,  $\mathcal{M} = \mathcal{B}_{[0,1]}$  denote the Borel  $\sigma$ -algebra, and  $\mathcal{N} = 2^{\mathbb{R}}$ . Let  $\mu$  Lebesgue measure on  $\mathcal{M}$ , and let  $\nu$  counting measure on  $\mathcal{N}$ . Prove that  $\Delta = \{(x, x) : x \in [0, 1]\} \subset [0, 1] \times [0, 1]$  is an element of the product  $\sigma$ -algebra  $\mathcal{M} \otimes \mathcal{N}$  and compute

$$\int_X \left( \int_Y \mathbf{1}_\Delta(x, y) d\nu(y) \right) d\mu(x), \quad \int_Y \left( \int_X \mathbf{1}_\Delta(x, y) d\mu(x) \right) d\nu(y). \quad (1)$$

Explain the relation to Tonelli's Theorem on product integration.

- (4) State the Hahn Decomposition Theorem. Prove, if  $\rho$  is a signed measure on the measurable space  $(X, \mathcal{M})$ , then there exist unique positive measures  $\rho_\pm$  on  $\mathcal{M}$  such that  $\rho_+ \perp \rho_-$  and  $\rho = \rho_+ - \rho_-$ .

Assuming  $\mu$  is a positive measure on  $\mathcal{M}$  and  $f \in L^1(\mu)$  is real-valued, identify  $\rho_\pm$  for the signed measure

$$\rho(E) = \int_E f d\mu.$$

## Part B

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5. State the Baire Category Theorem. Suppose  $X$  is a Banach space. Prove that, as a vector space,  $X$  does not have a countable basis. (Here countable means equinumerous with  $\mathbb{N}$ .)
6. Let  $(\varphi_n)$  be a sequence from  $L^1(\mathbb{R})$  with the following properties:
- (i)  $\varphi_n \geq 0$  for all  $n$ ,
  - (ii)  $\int \varphi_n dx = 1$  for all  $n$ ;
  - (iii) for each  $\delta > 0$ ,  $\lim_{n \rightarrow \infty} \int_{|x| > \delta} \varphi_n dx = 0$ .

Show, if  $f \in L^1(\mathbb{R})$ , then  $\varphi_n \star f$  converges to  $f$  in  $L^1(\mathbb{R})$ .

7. Let  $X$  be a Banach space,  $Y \subset X$  a closed subspace. Say  $Y$  is *complemented* in  $X$  if there exists a *closed* subspace  $Z \subset X$  such that  $Y \cap Z = \{0\}$  and  $Y + Z = X$ .

Prove that if  $Y$  is complemented in  $X$ , then there exists a bounded linear operator  $T : X \rightarrow Y$  such that  $T(y) = y$  for all  $y \in Y$ .

8. Let  $S : \ell^2(\mathbb{N}) \rightarrow \ell^2(\mathbb{N})$  denote the *shift operator* defined, for  $f = (f(n))_{n=0}^\infty \in \ell^2(\mathbb{N})$ , by

$$Sf(n) = \begin{cases} f(n-1) & n \geq 1; \\ 0 & n = 0. \end{cases}$$

Show  $S$  is an isometry (and in particular is bounded) and find  $S^*$  and  $SS^*$ . Is  $S$  unitary?